



SHOPRITE

FRESHCHAIN

Cold-chain rescue and protected-value control on SAP BTP

FreshChain connects store telemetry, stock ledger context, AI risk scoring, guided rescue actions, and financial proof in one governed operating process.

Collaboration: retail solutions SA and Shoprite

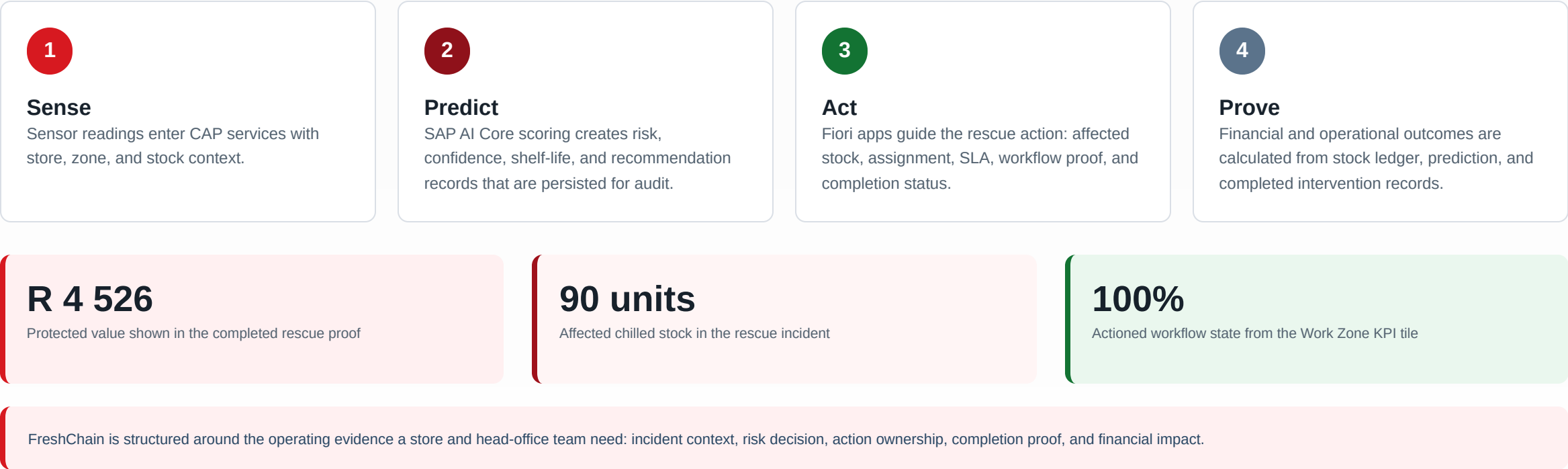
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Cold-chain rescue
SAP BTP product overview

Product process overview
FreshChain Fiori suite and SAP BTP services

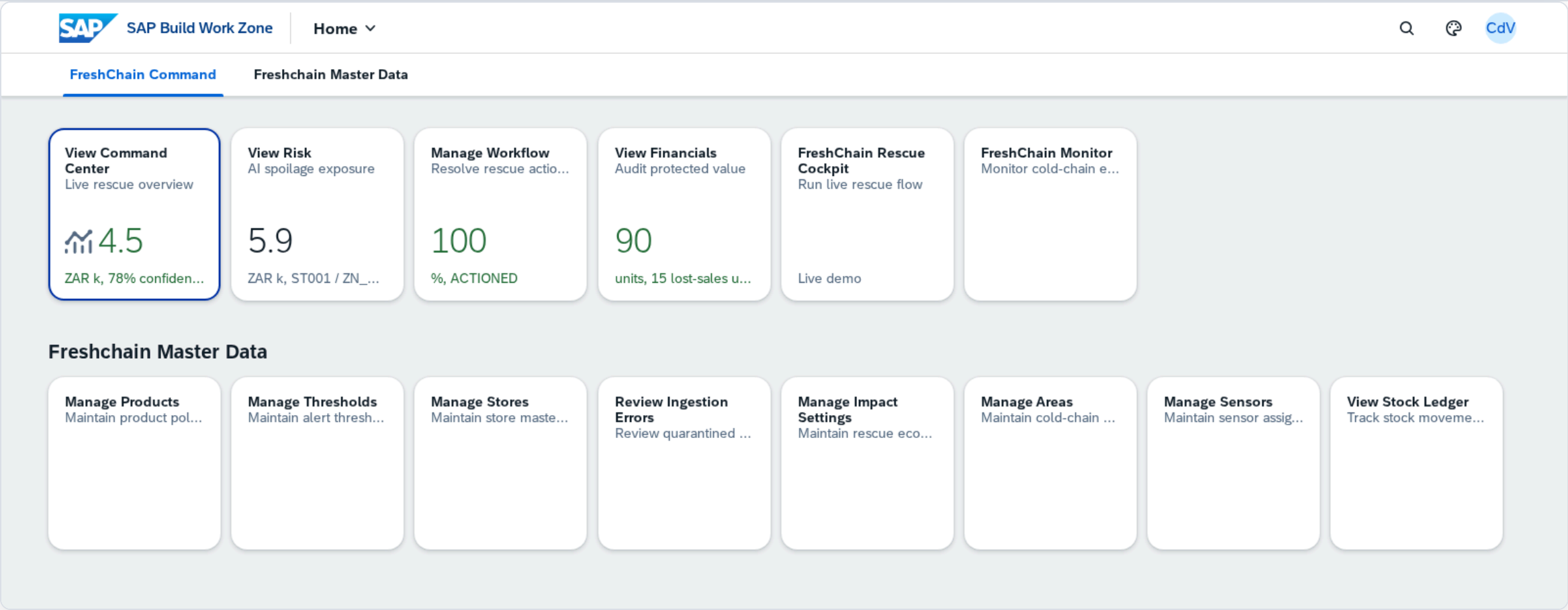
FreshChain cold-chain rescue process

A store receives a cold-chain signal, FreshChain calculates risk from live operational data, the store team executes a rescue action, and the outcome is recorded as protected value.



Work Zone home: operational tiles and KPI context

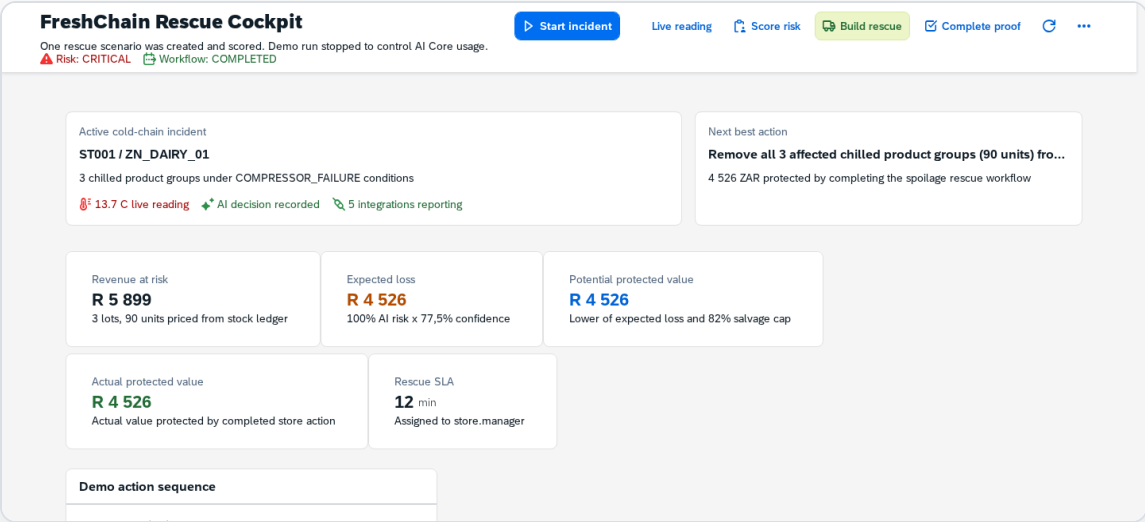
The Work Zone page separates command, risk, workflow, financial proof, rescue execution, monitoring, and master-data control into role-based entry points.



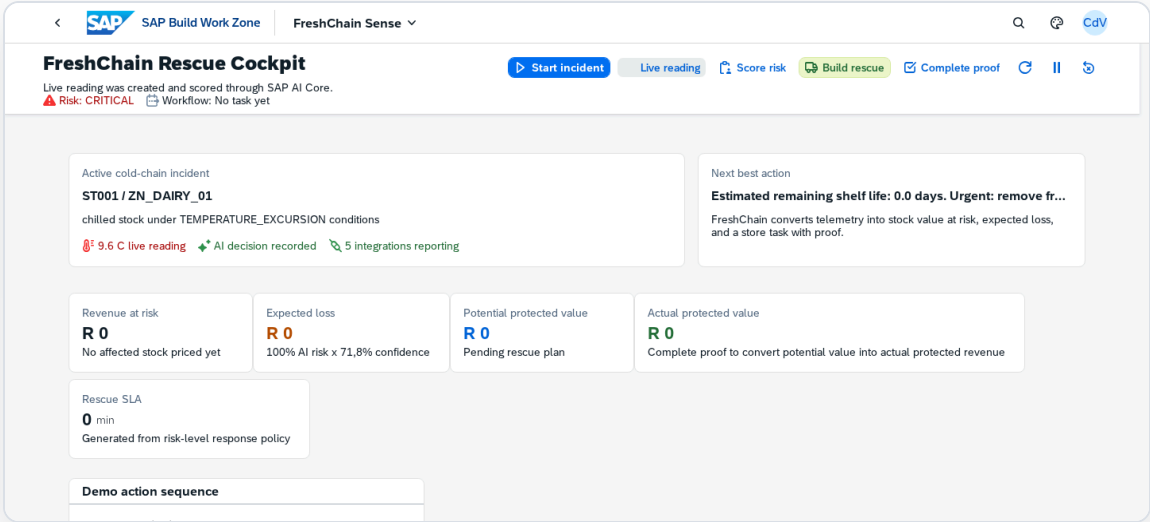
The command row covers protected value, exposure, rescue workflow, financial audit, incident execution, and cold-chain monitoring. The master-data row maintains the governed inputs behind the process.

FreshChain Rescue Cockpit

The Rescue Cockpit controls incident creation, reading capture, AI scoring, rescue planning, and proof completion from one action-oriented screen.



Start incident establishes the business context for the cold-chain exception.



Live reading records the condition that drives the downstream prediction and rescue action.

The reading is the trigger. Persistence, scoring, intervention planning, financial calculation, and proof are produced by FreshChain services and stored business records.

AI risk decision: score, confidence, recommendation

The prediction step converts the reading and stock context into a persisted decision record with risk level, confidence, remaining shelf-life, and recommended action.

<

 SAP Build Work Zone

FreshChain Sense ▾

🔍





FreshChain Rescue Cockpit

▶ Start incident

Live reading

 Score risk

 Build rescue

 Complete proof







Latest reading was scored through SAP AI Core.

 Risk: CRITICAL

 Workflow: No task yet

Active cold-chain incident

ST001 / ZN_DAIRY_01

chilled stock under TEMPERATURE_EXCURSION conditions

 9.6 C live reading

 AI decision recorded

 5 integrations reporting

Next best action

Estimated remaining shelf life: 0.0 days. Urgent: remove fr...

FreshChain converts telemetry into stock value at risk, expected loss, and a store task with proof.

Revenue at risk

R 0

No affected stock priced yet

Expected loss

R 0

100% AI risk x 71,8% confidence

Potential protected value

R 0

Pending rescue plan

Actual protected value

R 0

Complete proof to convert potential value into actual protected revenue

Rescue SLA

0 min

Risk

Critical spoilage exposure is linked to the affected store, zone, and stock lot.

Confidence

The score and confidence are persisted with the inference request for auditability.

Action

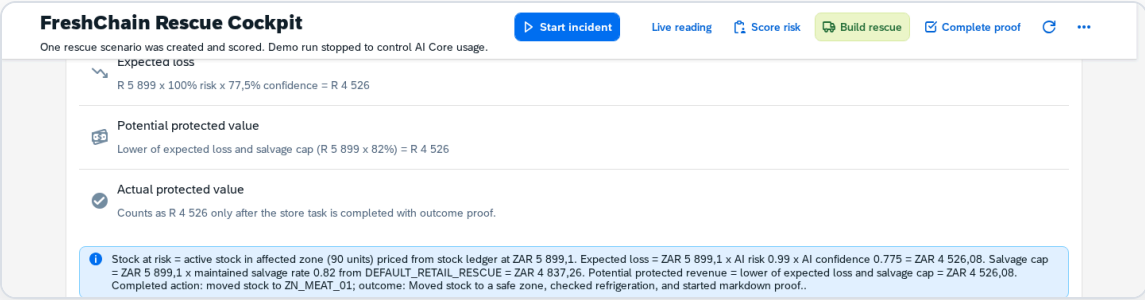
The recommendation explains what the store team should do next.

Trace

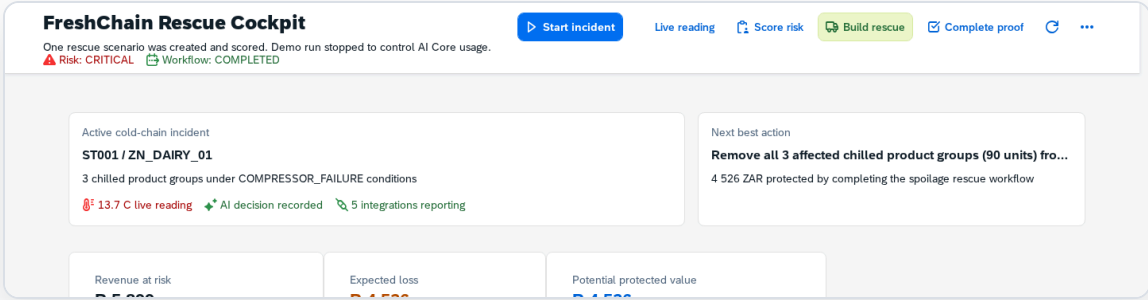
The decision remains linked to the rescue plan and later financial impact.

Store task completion and operational proof

The completed state records intervention outcome, workflow task reference, protected value, and operational instruction.



The task view identifies ownership and the required rescue action.



The completion view records outcome, protected value, and operational status.

RESCUED

Outcome status after completing the workflow proof

Task ID

The screen includes a workflow proof identifier for audit discussion

SLA

Action required within the operational time window

FreshChain Fiori suite

The application areas separate command, prediction, action, proof, monitoring, and integration evidence while keeping them on the same operating process.

COMMAND

FreshChain Command Center
Current KPI position for protected value, exposure, workflow action, and cold-chain status.

PREDICT

FreshChain Predict
Lists SAP AI Core risk decisions with score, confidence, store, zone, and prediction type.

ACT

FreshChain Act
Supports operational decisioning for stock rescue, movement, markdown, and store-level response.

PROVE

FreshChain Prove
Audits completed interventions and explains where protected revenue and outcome values come from.

MONITOR

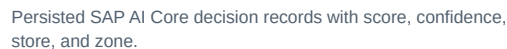
FreshChain Monitor
Tracks cold-chain events, ingestion health, and the operational telemetry needed to maintain confidence.

INTEGRATIONS

FreshChain Integrations
Platform hand-offs, AI Core proof, message processing, and runtime status.

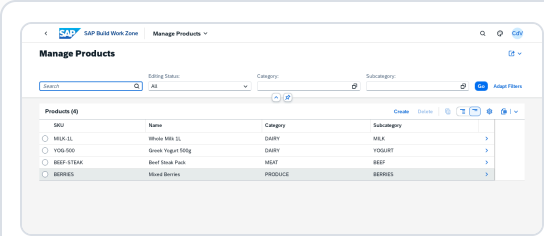
FreshChain keeps the core operating chain visible: command context, incident control, AI decision, rescue plan, completion proof, and financial outcome.

These applications persist the evidence created by the cockpit: interventions, AI decisions, protected-value proof, and cold-chain monitoring.



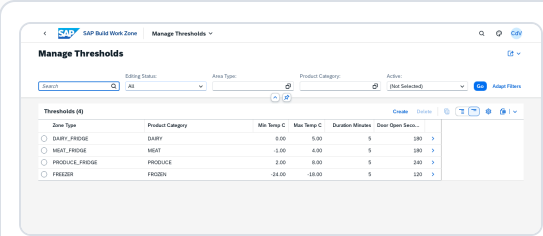
Supporting apps that make the process governed

The master-data apps govern the inputs used by rescue decisions, financial calculations, and operational monitoring.



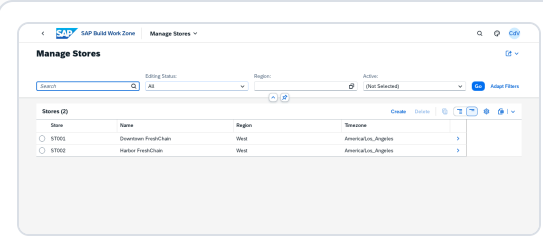
Manage Products

Product policy, category, and perishable stock attributes.



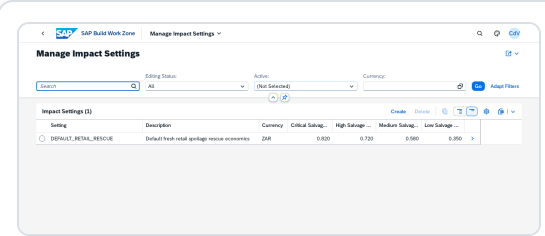
Manage Thresholds

Alert thresholds and cold-chain risk parameters.



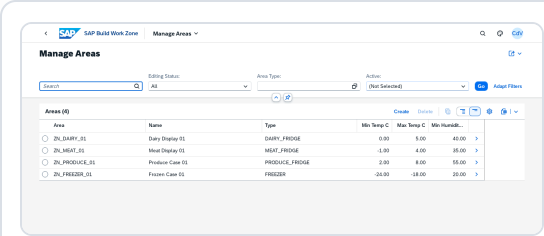
Manage Stores

Store master data and regional context.



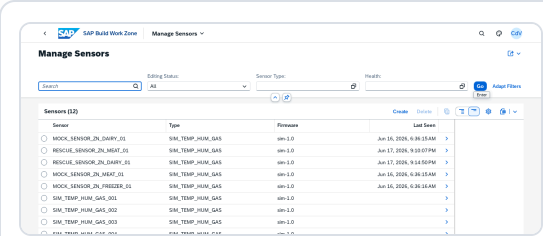
Manage Impact Settings

Economics used for intervention value calculations.



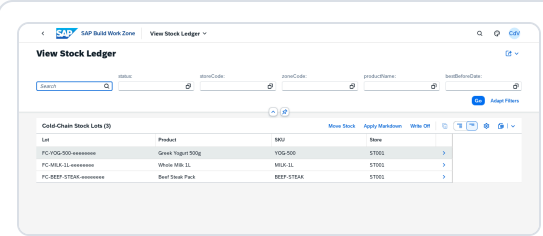
Manage Areas

Cold-chain zones and store-area definitions.



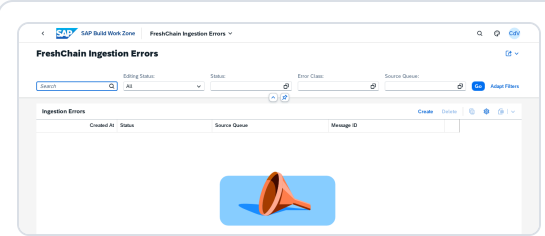
Manage Sensors

Sensor assignment, status, and telemetry source control.



View Stock Ledger

Stock movements used by risk and protected-value calculations.

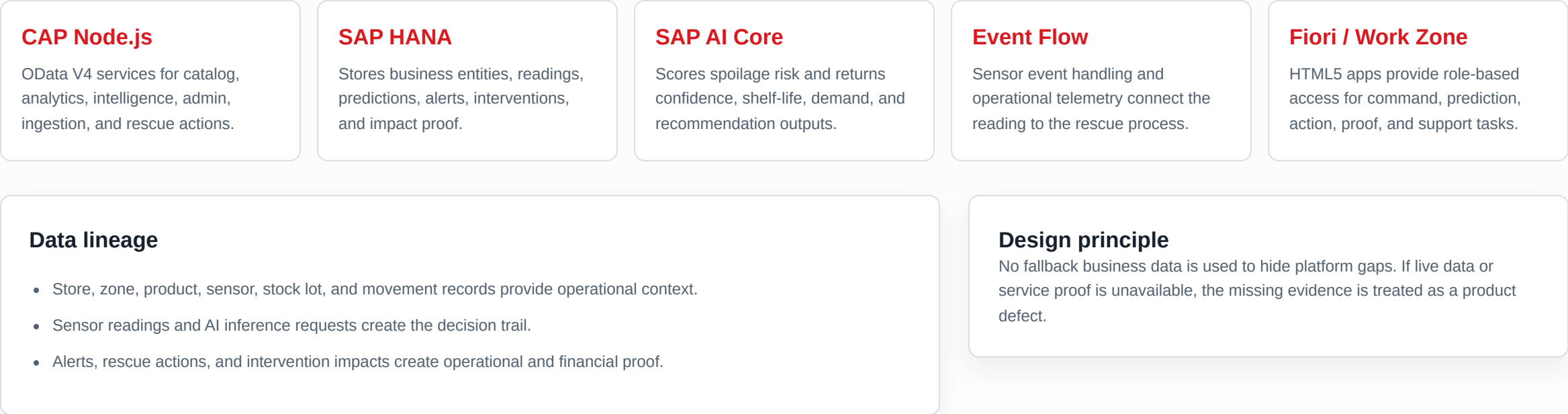


Review Ingestion Errors

Quarantined payloads and replay control for operations.

How SAP BTP services fit together

FreshChain is implemented as a CAP-backed Fiori suite on SAP BTP. Data, scoring, action, and proof remain in governed services rather than isolated screens.



FreshChain operating sequence

FreshChain handles the cold-chain rescue lifecycle from exception detection through protected-value proof.

Process sequence

- Detect a cold-chain exception from the store telemetry stream.
- Persist the reading with store, zone, product, sensor, and stock context.
- Score spoilage risk through SAP AI Core and store the decision record.
- Build a rescue plan from affected stock, recommendation, SLA, and rescue economics.
- Execute the store action and record workflow completion proof.
- Calculate protected value from stock ledger, risk, confidence, and salvage cap.
- Monitor recurring risk, ingestion quality, and intervention outcomes.

Product outcome

The closed loop gives store and head-office teams a clear record of what happened, why the action was recommended, who completed it, and how protected value was calculated.

Master data, thresholds, sensors, impact settings, and stock movements remain visible because they control the quality of the rescue decision.